MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FIFTH SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

MODEL QUESTION PAPER - THEORY

STATISTICS

MG5DSESTA300 – INTRODUCTION TO MULTIVARIATE ANALYSIS

Duration: 1.5 Hours **Maximum: 50 Marks**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer questions

Answer any seven questions. Each question carries **2 marks**.

1. How to use Hotelling′s T^2 statistic to test a given mean vector. [U] [1]
2. "T^2 is invariant under non−singular transformation" Verify. [U] [1]
3. Define Canonical Correlation Analysis in multivariate statistics. [K][1]
4. What are the three basic questions considered in profile analysis? [K][1]
5. When it is preferred to consider correlation matrix over covariance matrix in the case of factor analysis. [U][3]
6. Define Confirmatory Factor Analysis. [K][3]
7. State how EFA helps to identify unobserved constructs. [K][3]
8. List any two characteristics of orthogonal factor model. [K][3]
9. Under what condition do we obtain a quadratic discriminant function? [U][4]
10. What is a hyperplane in SVM? [K][4]

Part B

Short Essay questions

Answer any four questions. Each question carries **6 marks.**

11. Prove that the extraction of principal components from a dispersion matrix is the study of characteristic roots and vectors of the same matrix. [A][2]
12. Discuss the core structural characteristics of EFA. [U][3]
13. Compare and contrast Varimax and Equamax rotation methods in Exploratory Factor Analysis. [U][3]
14. Explain the geometric interpretation of factor rotation using factor axes. [U][3]
15. Explain the Maximum Likelihood method of estimating factor loadings. [U][3]
16. Explain the agglomerative approach to hierarchical clustering. [U][4]

Part C

Essay questions

Answer any one question. Each question carries **12 marks**.

17. Derive the connection between Hotelling′s T^2 statistic and Likelihood Ratio Test(LRT) for testing H0: $\mu=\mu 0$ in a p-variate normal distribution. [U][1]
18. Discuss the steps involved in performing correspondence analysis. [U][5]